MIL-I-14729A(MU)
5 June 1969
Superseding
MIL-I-14729(MU)
10 May 1968

MILITARY SPECIFICATION

INCREMENT, PROPELLANT, M90E1

1. SCOPE

1.1 This specification covers one type of propellant increment designated as M90E1 for Mortar Cartridge 81MM.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

MILITARY

MIL-B-117	-Bag, Sleeve and Tubing - Interior Packaging
MIL-B-2427	-Boxes, Ammunition Packing; Wood, Nailed
MIL-A-2550	-Ammunition & Special Weapons, General Specification for
MIL-B-3106	-Board, Composition, Water-Resistant, Solid
MIL-I-45607	-Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-652	-Propellant, Solid for Cannon Requirements and Packing
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

ARMY

20-4-619	- Marking Diagram and Sealing of Fiberboard Cartons for shipment of Propellant Increment
7548645	- Carton, Packing, Reusable Collapsible for High Explosives, Assembly Details, Packing and Marking
8796522	- Marking Diagram and Sealing for Wood Packing Boxes
8858577	- Marking Diagram and Sealing of Container, Metal, Universal M25 for Shipment of Propellants
8858848	- Marking Diagram and Sealing of Metal Lined Wooden Packing Boxes for Shipment of Propellants
9233369	- Charge A, Propellant Increment, M90E1 Mortar Cartridge 81MM
9233368	- A Bag, Propellant
9233371	- Charge B, Propellant Increment, M90E1, Mortar Cartridge 81MM
9233370	- B Bag, Propellant

PUBLICATIONS

ARMY

ET - 9233369 and 9233371 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

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(Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-90 (latest revision)".

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Increments.-The increments shall comply with all requirements specified on Drawing (dwg.) 9233368, 9233369, 9233370 and 9233371 and with all requirements specified in applicable specifications.

3.3 Strength of heat seal.-The specimen shall withstand application of a static load of 2.5 pounds per lineal inch, minimum, without peel-back of the heat seal and without tensile failure of the area adjacent the heat seal when tested as specified in 4.3.1 and 4.3.1.1.

3.4 Gross weight of propellant increment.-The weight of the loaded propellant increment within each lot shall not vary more than plus four or minus two grains from the established increment weight (assessed charge weight plus bag weight tare) for that lot, (see para. 4.2.3.1).

3.5 Workmanship.-All increments shall be free of dirt grease and other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

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a. Laminate from not more than one manufacturer.
b. Propellant from not more than one lot.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Increment (see dwg. 9233368, 9233369, 9233370 and 9233371)

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Bag or bag seam damaged to extent that propellant can escape	Visual	01001
Major:	AQL 0.40 percent		
101.	Bag or bag seam damaged but not to extent that propellant can escape or heat seal incomplete along any seam ...	Visual	01002
Minor:	AQL 2.5 percent		
201.	Length between attachment holes	Gage	01003
202.	Total length of increment ...	Gage	01004
203.	Location of attachment holes.	Gage	01005

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204.	Length of thread end of transverse seam incorrect ..	Visual-Gage	01006
205.	Stitches broken or missing .	Visual	01007
206.	Evidence of poor workmanship (see 3.5)	Visual	01008
207.	Marking illegible or incomplete	Visual	01009

4.2.2.2 Sealed inner liner (see 5.1.1, 5.1.2 or 5.1.3)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Liner damaged so that contents are exposed or liable to become exposed ...	Visual	02001
102.	Bag improperly sealed	Visual	02002
Minor:	None defined.		

4.2.3 Sealed wood packing box (see 5.1.1, 5.1.2 or 5.1.3)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Strapping missing, broken or loose	Visual-Manual	03001
102.	Board broken or split	Visual	03002
103.	Contents exposed	Visual	03003
Minor:	AQL 1.5 percent		
201.	Strapping improperly engaged	Visual-Manual	03004
202.	Contents move when shaken ..	Manual	03005
203.	Marking misleading, missing or unidentifiable	Visual	03006

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4.2.3 Testing

4.2.3.1 Gross weight of propellant increment; Under minimum, Critical defect - Code No. 04001; Over maximum, Major defect - Code No. 04002 (see 3.4).-The propellant increment gross weight shall be determined 100 percent and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weight determination, and if performed manually shall be performed by a different operator. Any propellant increment which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.2 Weight of propellant; Under minimum, Critical defect - Code No. 05001; Over maximum, Major defect - Code No. 05002 (see dwg. 9233369 and 9233371).-A sample of 315 propellant increment bags shall be randomly selected from each lot and the propellant shall be weighed independent of the bag. If one or more samples are less than the minimum allowable weight or if three or more samples exceed the maximum allowable weight, the lot shall be rejected.

4.2.3.3 Strength of heat seal (see 3.3) - Major defect, Code No. 06001.-Twenty sample bags shall be selected from each lot for this test, if any sample fails to comply with the applicable requirement, the lot shall be rejected. Testing shall be performed as specified in 4.3.1.

4.2.3.4 Air pressure test (see dwg. 8858848) - Major defect - Code No. 08001.-The containers shall be tested 100 percent. If any container fails to meet the requirement it shall be rejected and removed from the lot. Testing shall be performed as specified in 4.3.1.2 using equipment in accordance with 4.2.4.

4.2.4 Inspection Equipment - Equipment Tabulation Numbers ET-9233369 and 9233371 identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at ovrhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.2.4.2 Government use of contractor's inspection equipment. The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Conditioning.-The test samples shall be conditioned prior to testing at a temperature of 70 to 80 degrees Fahrenheit (°F) and shall be tested under these conditions unless otherwise specified.

4.3.1.1 Strength of heat seal.-One specimen, one inch wide, shall be cut perpendicular to the line of a seam of each sample bag. Location of the sample shall be random so that both end seams and long seams are sampled. One leg of the specimen shall be clamped so that the specimen hangs freely. The required load (see 3.3) shall be applied to the free end in such a manner as to avoid impact loading the seam. The load shall be allowed to act for one minute after which the seam shall be examined for peel-back of the heat seal. In the event that a one inch sample cannot be taken from an end seam the required load shall be reduced in proportion to the reduction in sample width.

4.3.1.2 Air pressure test for containers (see dwg. 8858848). The container shall be subjected to an internal pressure of 1/2 to 1 pound per square inch. A water manometer shall be assembled in the system. A drop of 0.7 inches or more, on the manometer in 15 seconds shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.1)

5.1.1 Level A.-Propellant increments shall be packed in heat sealed barrier bags complying with Type I, Class E of specification MIL-B-117. The barrier bag size shall be 22 1/2 inches long by 22 1/2 inches wide, nominal. Approximately 15 pounds of propellant shall be packed in each bag and closed by means of heat sealing. Two heat sealed bags shall be packed in a wooden box conforming to Grade A, Type I, Class 3 of specification MIL-B-2427. Both ends, sides, top and bottom of the wooden box shall be lined with 1/16 inch thick composition

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board conforming to Type I or II of specification MIL-B-3106. The box shall be closed and strapped in accordance with the appendix of the box specification. The net weight shall not exceed 30 pounds of propellant.

5.1.2 Level B.-Propellant increments shall be packed in accordance with packing provisions of MIL-STD-652.

5.1.3 Level C.-Propellant increments shall be packed in a fiberboard carton conforming to requirements for carton No. 1 of Drawing 7548645 except that liner shall be manufactured from barrier material conforming to Type I, Class E of specification MIL-B-117. Alternatively, a wood box meeting the requirements of Code of Federal Regulation Parts 171-179 may be used in lieu of the fiberboard carton.

5.2 Marking

5.2.1 Level A.-Exterior containers shall be marked in accordance with drawing 8796522. The dot marking shall be "Propellant Explosive Class A".

5.2.2 Level B.-The M17, M18 and M24 containers shall be marked in accordance with dwg. 8858848. The M25 container shall be marked in accordance with dwg. 8858577.

5.2.3 Level C.-The fiberboard carton or other container shall be marked in accordance with drawing 20-4-619.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Inspection equipment lists IEL.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further

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calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1375-A-047

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

DIRECT GOVERNMENT CONTRACT | SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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